**COMPUTING & INFORMATION SCIENCE DEPARTMENT**
**SCO400_SIT400_SCO444 – PROJECT**
Concept Paper

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| **Name:** | | **Reg No:** | |
| **Email Address:** | | **Telephone:** | |

## Proposed Project Title:

**QueueSmart Kenya: A Web-Based Digital Queue Management System for Huduma Centre**

## Problem Description and Justification:

**Long waiting times and inefficient queue management can reduce the quality and efficiency of services in Huduma Centres in Kenya. Customers often have to remain physically at service locations while waiting for their turn.**

## Proposed Solution:

**QueueSmart Kenya will allow users to book, receive digital queue numbers, monitor their position remotely, and receive notifications when their turn approaches. Staff will manage queues through a dashboard, while the system will estimate waiting times using current queue and service information.**

## Proposed Tools for Solution:

- **Frontend: React.js**
- **Backend: Python FastAPI**
- **Database: PostgreSQL**
- **Real-time communication: WebSockets**
- **Authentication: OTP**
- **Prediction: Python**